KATHIR V

Data Analyst | Python, SQL, Healthcare & Biological Data

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PROFESSIONAL SUMMARY

Fresher data analyst (B.Tech Bioinformatics, 2026) with project experience in Python, Pandas, NumPy and Excel on healthcare, image and biological datasets of up to 11,620 records. Skilled in data cleaning, feature engineering, model evaluation and visualization of results, with a focus on healthcare data.

TECHNICAL SKILLS

Languages: Python, SQL (basic), R

Data Analysis: Pandas, NumPy, Data Cleaning, Data Preprocessing, Feature Engineering, Exploratory Data Analysis, Data Visualization, Model Evaluation (accuracy, F1-score, confusion matrix)

Machine Learning: scikit-learn, XGBoost, Random Forest, TensorFlow, PyTorch

Tools: Excel, Jupyter Notebook, GitHub, Librosa, Biopython

PROJECTS

Physicochemical-Enhanced Embeddings for Protein Family Classification (Team Project) | Python, Biopython, scikit-learn

- Raised accuracy from 16.95% (standard 3-mer baseline) to 64.63%, while cutting features from 1,000 to 261 (74% fewer), by encoding sequences as 3-mers of hydropathy-grouped amino acids on 11,620 UniProt/Swiss-Prot sequences across 10 families.
- Implemented and compared Random Forest, SVM, KNN and Extra Trees, evaluated on an 80:20 stratified split using accuracy, F1-score and confusion matrices.
- Created performance visualizations and optimized the modelling workflow.

Safe Breath: Lung Sound Data Analysis & Classification | Python, Pandas, Librosa, scikit-learn | Google Play: play.google.com/store/apps/details?id=com.gs.safebreath

- Reached 91.59% accuracy across 14 respiratory-condition categories, by cleaning and preprocessing 3,530 audio recordings (denoising, silence removal, normalization, segmentation) and benchmarking 5 models.
- Built a 146-feature dataset (MFCC, spectral, ZCR, RMS), and balanced classes using augmentation and SMOTE+ENN (7,821 samples).
- Compared MLP, Random Forest, Extra Trees, XGBoost and an ensemble, using tuned hyperparameters and consistent evaluation metrics.

Blood Cell Image Analysis | Python, PyTorch, Score-CAM

- Reached 99.18% test accuracy on 4,886 BCCD cell images (RBC 4,153, WBC 372, platelets 361), by preprocessing, cropping and augmenting images and using a 70:15:15 split.
- Explained model results visually, using Score-CAM heatmaps.

EDUCATION

B.Tech, Bioinformatics | SASTRA Deemed University, Thanjavur | Aug 2022 – May 2026 | CGPA: 7.46/10

CERTIFICATIONS

- AI & Machine Learning and Data Science – Udemy, 2024
- Generative AI & Prompt Engineering – HCLTech, 2025